

RN00275

for S32G3 HSE Firmware 0.2.64.0

Rev. 1.0 — 20 March 2025

Release notes
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Document information

Information	Content
Keywords	HSE, S32G3, RN00275
Abstract	This release note is specific to HSE firmware for the S32G3 devices and describes the changes with each update of the HSE firmware along with any known issues for each corresponding version.



1 Getting Started

IMPORTANT NOTES:

This is the Standard Package variant of the HSE Firmware for S32G3.

1.1 Package content

This package contains the NXP S32G3 HSE Firmware 0.2.64.0:

- HSE Firmware (encrypted binary):
 - **Rev 1.1** (latest version of the SoC)
- HSE Firmware interface files
- HSE Service API RM
- HSE Security Installation Guide
- HSE_FW_S32G3_0.2.64.0_ReleaseNotes.pdf – this file
- The `license.txt` EULA file and `ApacheLicense2.0.txt`
- The `uninstall.exe` utility for removing the HSE FW binary

Note:

The Demo Application is provided separately and contains details on how to provision HSE FW on virgin devices and demonstrates common use cases of its security features.

The following associated documentation can be obtained from the NXP website (<https://www.nxp.com>), Sign In / Register > My NXP Account > Secure Files:

- *HSE H/M Firmware Reference Manual VERSION 2.6*

1.2 Installation

Follow the installation steps in the demo application.

When targeting the usage of AUTOSAR software stack in the application, it is recommended to install also the RTD crypto driver from the AUTOSAR RTD package for the targeted platform by following its installer steps.

2 Release Specifics

2.1 Release Details

This is the HSE Firmware 0.2.64.0 release for the S32G3 platforms.

This release is a Service Release and has RFP (RTM) quality in terms of testing and quality documentation. Service Release contains all features and is fully tested on Customer Engineering Samples or Qual Intend Samples. RFP (RTM) release is software release that can be used in cars production. RFP (RTM) release is delivered with complete software documentation and quality package.

The provided example code shows how to set up and use the HSE FW and to perform basic crypto operations (refer to the documentation that comes with the demo application). The examples show how to:

- Boot the demo-application (secure mode)
- Install the Firmware
- Load the key(s)
- Perform cryptographic operations

This release was developed and tested using:

- Chip: **P32G399AAK1CVUC (rev 1.1)**
- Design Board: **S32G-VNP-EVB**

Standard HSE Firmware package contains the following configuration:

- 20 RAM keys, 40 NVM symmetric keys, 12 NVM asymmetric keys
- 8 SMR entries, 4 CR entries
- Support only for ECC-256 bits and CURVE25519 curve (Montgomery and Twisted)
- Maximum key size limitations: HMAC - 512 bits, ECC - max 256 bits, RSA- max 2048 bits
- SHA3, IPSec, Classic DH and Burmester-Desmedt services are not supported
- CURVE448 is not supported (Montgomery and Twisted)

Note:

- *The release package provides the firmware image for rev 1.1 of the S32G3XX chip.*
- *For the Premium Package variant, the customers would need to purchase “premium S32G3 security parts” (to run Premium Package variant in production).*

Important: *Ensure that the proper SoC revision and the latest HSE Firmware version are used.*

Implemented Errata:

- **ERR051655** erratum - OCOTP: Incorrect data may be read from fuses after software power down and up of the Fusebox.
- **ERR052243** erratum - TMU: Unexpected temperature readings during monitoring mode.

2.2 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP:

- S32G399A

2.3 Device Bricking scenario for HSE Firmware

- N/A

2.4 Security Aspects

The current release of HSE Firmware implements countermeasures protecting against remote and local software attacks.

3 Known Issues

- N/A

4 Changes List

4.1 Changes in 0.2.64.0

Added

- Add `HSE_DISABLE_KEY_USAGE_ATTR_ID` attribute used to disable a key until next reset.

Updated

- Removed support for `hseTempSensVioConfig_t` attribute due to erratum **ERR052243**; the HSE reaction to temperature violation can no longer be configured (an `HSE_SRV_RSP_INVALID_PARAM` status is returned).
- Allow only one SHE RAM key slot to be configured. The SHE RAM group should be mapped to index zero.
- For the `hseSignSrv_t` service, if the `bInputIsHashed` parameter is set to `TRUE` and the `hashAlgo` for the `signScheme` scheme is provided, HSE checks that the `inputLength` is equal to the hash output length.
- Enable SHA384 and SHA512 support for `hseKeyVerifySrv_t` service.

Fixed

- SYS-IMG size is not correct when the IVT flash page size is higher than 4kB.
- HSE does not issue a functional reset when `hseAttrOtpBootSeq_t` is set (`BOOT_SEQ` bit in fuses is programmed) and `IVT_BOOT_SEQ` is zero. HSE only goes to shut down.
- Signing an AppBL image (`hseBootDataImageSignSrv_t`) while having the MU memory regions configured returns `HSE_SRV_RSP_INVALID_ADDRESS` if the destination memory range does not fit within a configured memory region.

4.2 Changes in 0.2.51.0

Added

- Extended the `disableOtpRollbackProtect` parameter of `hseOtpRollbackProtectionPolicy_t` attribute to include the `HSE_ON_DEMAND_ANTI_RBC_UPDATE` option: the rollback protection mechanism is enabled, and the anti-rollback counter can be updated in fuses only on-demand by calling the `hseOnDemandAntiRbcUpdateSrv_t` service.
- Added the `hseOnDemandAntiRbcUpdateSrv_t` service that can be used to update on demand the anti-rollback counter in fuse (of SYS-IMG or/and FW IMG). This service can be used only if `disableOtpRollbackProtect == HSE_ON_DEMAND_ANTI_RBC_UPDATE` (refer to `hseOtpRollbackProtectionPolicy_t` attribute).
- Added `hseFirmwareVerifySrv_t` service to verify on demand the blue or pink firmware image (in external flash or SRAM).
- Added the `hseAttrOtpBootSeq_t` attribute (`OTP_BOOT_SEQ`) to enforce the `BOOT_SEQ` flag in IVT not to be changed to 0 in secure boot mode. If `IVT_BOOT_SEQ == 1` (secure boot), this attribute can be used to set the `in OTP_BOOT_SEQ` in fuses. At start-up, HSE verifies the value of `OTP_BOOT_SEQ` (from fuses) against the value of `IVT_BOOT_SEQ` as follows: if (`LC == OEM_PROD` or `IN_FIELD`) and (`OTP_BOOT_SEQ == 1`) and (`OTP_BOOT_SEQ != IVT_BOOT_SEQ`), apply a functional reset; otherwise, continue the boot sequence.
- Added the read-only `hseRbCounterInfo_t` attribute to provide:
 - the OTP counter for SYS-IMG
 - the OTP counter for FW-IMG
 - the counter from SYS-IMG header (returns 0xff if the SYS-IMG was not loaded)
 - the counter from blue FW header (returns 0xff if the firmware was loaded from the pink FW image).
- Updated `hseKeyVerifySrv_t` service: added support to verify the asymmetric keys inside HSE key store (public key, key pair, or `_PUB_EXT` keys) providing a SHA256 or CMAC tag. Removed the SHA384 or SHA2_512 options.

- Added `HSE_SRV_RSP_RNG_INIT_IN_PROGRESS` service response: this response status can be received when the Get Random Number service is called and the RNG initialization is in progress. The application can try again later.
- Added the `hseDisablePairWiseConsistencyTest_t` attribute (SET-ONLY-ONCE-ATTR attribute) to disable the pair-wise consistency check when importing an ECC/RSA key pair to increase the key import performance. By default, HSE checks the pair-wise consistency.

Updated

- Enhanced security countermeasures (code hardening).
- For the `hseSignSrv_t` service, if the `bInputIsHashed` parameter is set to `TRUE` and the `hashAlgo` for the `signScheme` scheme is provided, HSE checks that the `inputLength` is equal to the hash output length.
- NXP ROM provision keys cannot be used to export keys.
- When rollback protection is enabled and status bit `HSE_WA_OTP_FUSE_WRITE_FAILURE_ON_BOOT` is set, the application should trigger a destructive reset or POR such that the HSE firmware retries programming the fuse.
- Reduced `HSE_MAX_NUM_OF_SGT_ENTRIES` to 16 entries (instead of 32).
- Reduced `MAX_STREAMING_CONTEXT_SIZE` to 372 bytes (instead of 616 bytes).
- Default `HSE_DEFAULT_MIN_FAST_CMACE_TAG_BITLEN` is 32 bits (instead of 64).
- Super User (SU) rights are no longer mandatory for storing the application specific data in persistent memory via `HSE_APP_SPECIFIC_DATA_ATTR_ID` attribute (refer to `hseAppSpecificData_t` structure).
- Improved the HSE Service API Reference Manual.

Fixed

- HSE entered shutdown mode with fatal error `GSR= 0x73080001`: when the RNG re-initialization is triggered due to a reseed failure (RNG reseeding can fail), it shall not be interrupted by high-priority crypto operation such as AES encryption.
- HSE entered shutdown mode with fatal error `GSR=0x33AA0001`: Fix the race condition when the watchdog idle/running state changes.
- The host cannot access the shared memory when HSE is in shutdown mode; make HSE-host shared memory accessible by host (even if HSE went to shutdown mode).
- The input length cannot zero for ECDSA.
- The second public key of the Burmester-Desmedt HSE service cannot be exported.
- Fix the boot sequence when the `VDD_EFUSE` is grounded and the FW or SYS_IMG are not updated.
- Do not allow monotonic counter read/increment if the counter table loading fails.

4.3 Change in 0.2.22.0

Added

- Added the firmware build information attribute (see `HSE_FW_BUILD_INFO_ATTR_ID`). This is a read-only attribute that provides a 8-byte unique ID, the date and time of the build.
- Added `HSE_DISABLE_APP_SPECIFIC_DATA_WRITE_ATTR_ID` attribute (see `hseDisableAppSpecificDataWrite_t` structure) which disables writing of application-specific data in SYS-IMG (disable writing of the `HSE_APP_SPECIFIC_DATA_ATTR_ID` attribute).
- An ECC private key can be imported without providing the public key. The public is computed internally from the private key.
- Indirect SYS-IMG access: the SYS-IMG pointer from IVT can point to a list of 16 addresses/entries (instead of pointing directly to the SYS-IMG location). A non-empty entry has the address value different from `0xFFFFFFFF`. At reset, HSE firmware parse the entry-list (from 0 to 15) to find the lasted non-empty entry, and uses the address found to read-out the SYS-IMG. In this way, the SYS-IMG can be written at multiple flash

locations. To enable this indirect access of the SYS-IMG, the IVT must be configured to include a marker at a specific offset in IVT (refer the HSE Firmware reference manual).

Updated

- Updated the fuse read/write handlers to cover **ERR051655** erratum. Additional code hardening was included.
Important: *Ensure that the latest HSE Firmware is used.*
- Enhanced security countermeasures (code hardening).
- Update the register address for `HSE_GPR_STATUS_ADDRESS`.
- Updated the `HSE_APP_SPECIFIC_DATA_ATTR_ID` attribute that allow to store application-specific data in SYS-IMG (persistent memory); refer to `hseAppSpecificData_t` structure.
- Updated the Publish SYS-IMG service (`hsePublishSysImageSrv_t`): the `HSE_PUBLISH_UPDATED_DATA_SET` option was removed.
- Allow Lifecycle advancement without Super User Rights.
- Increased the performance for EdDSA signature verification for big data.
- Enhanced the Firmware Update service: code hardening and added additional checks.
- Improved the HSE Service API Reference Manual (HSE header files comments).
- Increased the plaintext FW image size by 32KB. The size of the image as provided by NXP (pink image) is 389KB.

Fixed

- The first RSA signature verify costs 6x times than the same service requested after that.
- Incorrect AES-GCM tag is computed when IV counter wraps.
- An encrypted SMR install request from flash with a large SMR size cause HSE to go to shutdown mode.
- A self-test firmware integrity request without a prior publish sys-image request cause HSE to go to shutdown mode.
- Anti-rollback counter for the HSE FW blue image is incremented twice at first installation. Note that the rollback protection for HSE FW blue image can be disabled using an SYS-IMG attribute.
- Set the `HSE_FW_ROLLBACK_MARKER` marker in `BOOT_POR_CTRL_REG` register if at least one host core is booted.
- Removed the QSPI reads done by HSE at start-up after booting the application core(s).
- If the length of the shared secret slot is ≥ 256 bytes, the TLS 1.2 KDF pre-master computation using DHE-PSK schemes failed

4.4 Changes in 2.16.0

Added

- Support for boot data images (IVT/DCD/ST-DCD/AppBL for BSB) MAC generation/verification for the new HW revision – `hseBootDataImageSignSrv_t` and `hseBootDataImageVerifySrv_t` updated:
 - On previous revision (1.0), these services are using a static IV, HSE FW giving as output at signature generation only the GMAC.
 - On latest revision (1.1), when generating the signature, a random IV is generated by the HSE FW and returned to the application (along with the GMAC). This IV needs to be incorporated into the image at the fixed offset (as per SoC RM). When verifying the GMAC using the HSE FW service, the IV has to be already part of the image.
 - HSE FW supports these services for both revisions, however the generated artifacts are not interchangeable – i.e. an IVT authenticated for the previous revision cannot be used on the newer revision and vice versa.

- Revision can be read from SIUL2_0.MIDR1 register.
- Support for HSE FW released image for the new HW revision
 - The two encrypted binaries cannot be interchanged – i.e. the image for the previous revision (1.0) cannot be used on the newer revision and vice versa.
- Support for Extend Key Catalog service - `hseExtendKeyCatalogSrv_t`.
- Support for ROM public keys - `HSE_ROM_KEY_ECC256_PUB_KEY0`.
- Support for a new attribute by which application specific data can be stored in the SYS_IMG - `hseAppSpecificData_t`.
- Support for SGT input in CMAC with counter service - `hseCmacWithCounterSrv_t`.
- Support for SMR verification options when performing on-demand, run-time verification - `hseSmrVerificationOptions_t`.
- Support for SMR/CR entry erase - `hseSmrEntryEraseSrv_t` `hseCrEntryEraseSrv_t`.
- Support for handling data provided in the DDR0 (1.5G-2G) range for HSE descriptor, input for SipHash and input for Pure-EDDSA.

Updated

- Increased maximum number of regions configurable via the attribute `HSE_MEM_REGIONS_PROTECT_ATTR_ID` – from 6 to 12 regions per MU.
- Increased representation of the key counter for the NVM non-SHE keys - `hseKeyInfo_t` – from 28 bits to 32 bits.
- Minimum number of iterations for PBKDF2 – 100.
- Minimum length of IV used with the GMAC scheme for key import/export in an authenticated container – 12 bytes.
- Internal scheduler by increasing 5 times the time slot allocated for the fast jobs' execution.
- SMR periodic checks are disabled until the next reset if the associated SMR is updated.
- Enhanced HSE FW update service.
- Enhanced reactions to tamper violations.
- Enhanced security countermeasures.

Fixed

- Anti-rollback counter for the HSE FW blue image is incremented twice at first installation.
- TLS 1.2 KDF pre-master computation using DHE-PSK schemes and shared secret with length ≥ 256 bytes fails.
- When HSE FW is running from the backup location, after eight consecutive functional resets HES does not respond.
- When `BOOT_SEQ = 1` and HSE FW is running from primary location but fails before complete initialization (e.g. not application boots), the device will go in serial boot mode before attempting to run HSE FW from the backup location.

4.5 Changes in 0.21.0

Added

- NVM attribute for configuring the policy on HSE FW rollback protection - `hseOtpRollbackProtectionPolicy_t`. Can be used to enforce HSE FW to boot only from device-specific (blue image) and/or to disable the rollback protection altogether.
- NVM attribute to configure HSE reaction to a tamper violation – it can either go to shutdown or directly issue a reset - `hseResetSocOnTamper_t`.
- AAD support for encrypted SMR - `hseSmrDecrypt_t`.
- Support for compressed ECC keys - `hseEccKeyFormat_t`.

- Key verify service - `hseKeyVerifySrv_t` – verifies a hash/MAC (provided by the application) over a symmetric key from HSE key store.
- `HSE_KF_USAGE_XTS_TWEAK` usage flag that must be set for AES-XTS tweak keys.
- SGT support for AES-XTS service.
- `HSE_KF_USAGE_OTFAD_DECRYPT` usage flag that must be set for keys used for OTFAD decryption.
- AES block mode mask as part of the AES key info. It restricts the usage of an AES key only to those specified by `hseAesBlockModeMask_t` member. If set to 0 then no restrictions apply.
- The key derive copy restrictions on starting offset and number of bytes extracted - `hseKeyDeriveCopyKeySrv_t`.
- Support for different lengths of the iteration counter for SP800-108 and SP800-56C-TwoStep KDF schemes besides 32 bits; refer to `hseKdfSP800_108Scheme_t`.

Updated

- Increased HSE raw FW (plaintext FW) size by 32KB. The size of the image as provided by NXP (pink image) is 357KB.
- Policy on encrypted keys import/export. If encrypted, it must also be authenticated.
- TLS 1.2 pre-master secret generation and KDF - `hseKeyGenTls12RsaPreMaster_t` and `hseKdfTLS12PrfScheme_t`.
- OTFAD services by adding the instance number.
- Minimum MAC length to 16 bytes for plain SMR initial authentication proof, authenticated key import/export and system authorization services.
- Minimum MAC length to 8 bytes for MAC service.

Fixed

- HSE FW does not handle ECC errors in program images (i.e. SYS_IMG, SMR, AppBL in basic secure boot) when loading from external flash via the QSPI if the QSPI interrupt pin is connected to the external flash and this pin transitions low
 - HSE clears the ECC error bits in QSPI.FR until `HSE_STATUS_INIT_OK` status bit is set.
- HSE is non-operational when loading a valid SYS_IMG partially (`BOOT_SEQ=0`).
- HSE status and error bits are not updated on the application side (MU.FSR and MU.GSR) until `HSE_STATUS_INIT_OK` is set.
- ST-DCD cannot be signed/verified using the HSE FW boot data sign/verify services.
- The quality of random number generation not being ensured if `XBAR_CLK` is below 200MHz.

Removed

- MD5 support.
- SHA1 support for key derivation services.
- HMAC and CMAC support as PRF for NXP Generic KDF scheme.
- SipHash variant `HSE_SIPHASH_VARIANT_32` and the support of 64-bits SipHash keys.
- Specific fatal error events for tamper violations - `hseError_t`.

5 List of Limitations Existing in This Release

- The RNG re-initialization is triggered at the first call of the following services:
HSE_SRV_ID_GET_RANDOM_NUM, HSE_SRV_ID_KEY_GENERATE or HSE_SRV_ID_SYS_AUTH_REQ.
- VDD_EFUSE must be connected to 1.8V to write the HSE fuse area. If VDD_EFUSE is connected to GND, the HSE fuse area can only be read (cannot be written). The fuses are written:
 - By the application, through the Set Attribute service (e.g. life cycle, ADKP key, debug authorization method, etc.). In this case, the application must supply power (1.8V) to the VDD_EFUSE pin.
 - By HSE FW, at start-up when the blue FW-ING/SYS-IMG is updated and disableOtpRollbackProtect parameter is set to “NO” (refer to hseOtpRollbackProtectionPolicy_t attribute).
- During SD card booting, the first 36KB (start_address = 0x34000000, length = 0x9000) from SRAM are used. This memory region can be used by the application after a successful HSE initialization (HSE status shall be HSE_STATUS_INIT_OK for BOOT_SEQ = 0 and HSE_STATUS_BOOT_OK for BOOT_SEQ = 1).
- Defining SMR entries that are checked periodically may impact the overall HSE FW performance.
- The achieved code coverage can be found in the Quality Matrix summary report file that comes with this release. During this SW product development, the whole code went through a diligently inspection-based review and Static Code Analysis (MISRA, HIS).

6 List of Services Available

All available HSE features/services are also listed in the *hse_h_config.h* file (from HSE Interface). All other features not listed in the table below (or enabled in *hse_h_config.h* file) are **NOT supported**. All available services applicable for both configuration device.

Table 1. Available HSE Features/Services

Service Class	HSE Service ID	Description/Notes
Administrative	HSE_SRV_ID_SET_ATTR	Set an HSE attribute. Attributes related to FUSE memory can be written only once (e.g. Debug Key) or can only be advanced (e.g. Life cycle). Care must be taken.
	HSE_SRV_ID_GET_ATTR	Get an HSE attribute.
	HSE_SRV_ID_SELF_TEST	Performs a self-test on a specific security block or a full self-test. (HSE FW integrity, RNG, AES, HASH, MAC, CRC, RSA, ECC).
	HSE_SRV_ID_CANCEL	Cancel a one-pass or streaming service on a specific channel. An HSE service request can be cancelled if it is in the processing queue and NOT passed to the hardware to be executed.
	HSE_SRV_ID_FIRMWARE_UPDATE	HSE firmware update (generates the HSE FW blue image).
	HSE_SRV_ID_FIRMWARE_VERIFY	Verifies the blue or pink FW images (in SRAM or external flash).
	HSE_SRV_ID_SYS_AUTH_REQ	SYS Authorization request used to request the Super User rights (CUST/OEM).
	HSE_SRV_ID_SYS_AUTH_RESP	SYS Authorization response (response to SYS Authorization Request)
	HSE_SRV_ID_BOOT_DATA_IMAGE_SIGN	Generate the signature on IVT, DCD, and SELF TEST images. Also, signs the APP image for Basic Secure Boot (BSB).
	HSE_SRV_ID_BOOT_DATA_IMAGE_VERIFY	Verify the signature on IVT, DCD, and SELF TEST images. Also, verifies the APP image for Basic Secure Boot (BSB).
	HSE_SRV_ID_IMPORT_EXPORT_STREAM_CTX	Import and Export service for the crypto streaming context.
	HSE_SRV_ID_PUBLISH_SYS_IMAGE	Publish SYS-IMAGE file in System RAM.
	HSE_SRV_ID_GET_SYS_IMAGE_SIZE	Get SYS-IMAGE size
	HSE_SRV_ID_VERIFY_SYS_IMAGE	Verify SYS-IMAGE integrity after it is written in the external flash.
	HSE_SRV_ID_PUBLISH_LOAD_CNT_TBL	Request to publish/load the NVM container for the Monotonic Counter table
	HSE_SRV_ID_INSTALL_OTFAD_CTX	Install an On-The-Fly AES Decryption (OTFAD/IEE) context.
	HSE_SRV_ID_ACTIVATE_OTFAD_CTX	Activate on-demand OTFAD context.

Table 1. Available HSE Features/Services...continued

Service Class	HSE Service ID	Description/Notes
	HSE_SRV_ID_GET_OTFAD_CTX	Get OTFAD context information.
	HSE_SRV_ID_PREPARE_FOR_STANDBY	Prepare HSE before the system goes to Stand-by mode.
	HSE_SRV_ID_CONFIG_COUNTER	Configures the monotonic counters.
	HSE_SRV_ON_DEMAND_ANTI_RBC_UPDATE	Update on demand the anti-rollback counter(s) in fuses for SYS-IMG and/or FW-IMG.
Key Management	HSE_SRV_ID_LOAD_ECC_CURVE	Load the domain parameters for a Weierstrass ECC curve. This service can be used to support additional Weierstrass ECC curves (Not supported by default). The loaded ECC curve domain parameters are persistent.
	HSE_SRV_ID_FORMAT_KEY_CATALOGS	Format key application key catalogs (RAM & NVM).
	HSE_SRV_ID_ERASE_KEY	Erase NVM/RAM key(s). Erase key service depends on authorization rights. One or multiple keys can be erased. Selected key group can also be erased
	HSE_SRV_ID_GET_KEY_INFO	Get key properties (flags).
	HSE_SRV_ID_IMPORT_KEY	Import a key. Uses all algorithms supported by HSE firmware: • Plain form or AES / RSA encrypted. • MAC authenticated (refer to supported MAC algorithms) or RSA / ECDSA signed. • Import key restrictions depend on sys authorization rights. The restrictions are described by the service in the interface.
	HSE_SRV_ID_EXPORT_KEY	Export a key. Uses all algorithms supported by HSE firmware: • Plain form (only public keys) or AES / RSA encrypted. • MAC authenticated (refer to supported MAC algorithms) or RSA / ECDSA signed. • Export key restrictions depend on authorization rights. The restrictions are described by the service in the interface.
	HSE_SRV_ID_KEY_GENERATE	Request to generate a symmetric/asymmetric key. • Random symmetric key generation • RSA and ECC key pair generation
	HSE_SRV_ID_DH_COMPUTE_SHARED_SECRET	ECC Diffie-Hellman Compute Key (shared secret): • SEC curves: SECP256R1 • Brainpool curves: BRAINPOOLP256R1

Table 1. Available HSE Features/Services...continued

Service Class	HSE Service ID	Description/Notes
		- Montgomery curve: CURVE25519 1 user-defined ECC curves (see Load ECC curve service)
	HSE_SRV_ID_KEY_DERIVE	Perform a key derivation function: NXP Generic KDF, Extract KDF, SP800_56C One Step, SP800_56C Two Step, SP800_108 (Only Counter Mode), ANS_X963, ISO/IEC 18033 KDF2, ISO/IEC 18033 KDF1, PBKDF2HMAC, HKDF, IKEV2, TLS12 PRF
	HSE_SRV_ID_KEY_DERIVE_COPY	Extract a key from the derived key material to a key slot.
	HSE_SRV_ID_KEY_VERIFY	Verifies a provided hash/MAC over a symmetric/asymmetric key inside HSE key store.
	HSE_SRV_ID_EXTEND_KEY_CATALOG	Update the NVM or RAM key catalogs by appending new key groups.
	HSE_SRV_ID_SHE_LOAD_KEY	Load a SHE key using the SHE memory update protocol.
	HSE_SRV_ID_SHE_LOAD_PLAIN_KEY	Loads SHE specification-based RAM key as plain text.
	HSE_SRV_ID_SHE_EXPORT_RAM_KEY	Exports RAM key as per SHE specification.
	HSE_SRV_ID_SHE_GET_ID	GETUID as per SHE specification.
	HSE_SRV_ID_SHE_BOOT_OK	The command is used to mark successful boot verification for later stages than CMD_SECURE_BOOT. For more details, see SHE specification
	HSE_SRV_ID_SHE_BOOT_FAILURE	The command will impose the same sanctions as if CMD_SECURE_BOOT would detect a failure but can be used during later stages of the boot process. For more details, see SHE specification.
ROM Keys	N/A	Support for ROM keys (without RSA).
Cryptographic	HSE_SRV_ID_HASH	Hash service (one-pass and streaming): - SHA1 - SHA-224, SHA-256, SHA-384, SHA-512, SHA-512/224, SHA-512/256 - Miyaguchi-Preneel compression function (SHE specification support)
	HSE_SRV_ID_MAC	Request to generate/verify a Message Authentication Code (MAC): - AES-CMAC, AES-GMAC, AES-XCBC-MAC - HMAC_(SHA1, SHA_224, SHA_256, SHA_384, SHA_512)
	HSE_SRV_ID_FAST_CMACE	Low latency, high performance CMACE generate/verify request
	HSE_SRV_ID_SYM_CIPHER	Symmetric encryption/decryption (one-pass and streaming): AES-128/-192/-256: ECB, CBC, CTR, OFB, CFB
	HSE_SRV_ID_AEAD	AEAD encryption/decryption: AES-CCM-128/-192/-256 (one-pass, no streaming support)

Table 1. Available HSE Features/Services...continued

Service Class	HSE Service ID	Description/Notes
		AES-GCM-128/-192/-256 (one-pass and streaming)
	HSE_SRV_ID_XTS_AES_CIPHER	XTS AES encryption/decryption.
	HSE_SRV_ID_SIGN	Request a Digital Signature Generation/Verification (one-pass and streaming): - RSASAA_PSS (from 1024 up to 2048 bits key) - RSASAA_PKCS1-v1_5 (from 1024 up to 2048 bits key) - ECDSA (all supported ECC curves) - EDDSA (for ED25519 curve)
	HSE_SRV_ID_RSA_CIPHER	RSA encryption/decryption: - RSAAES-PKCS1-v1_5 (from 1024 up to 2048 bits key) - RSAAES-OEAP (from 1024 up to 2048 bits key)
	HSE_SRV_ID_AUTHENC	Combined Authenticated Encryption service: - AES_(ECB, CBC, CTR, CFB, OFB) -THEN- HMAC_ (SHA1, SHA2_224, SHA2_256, SHA2_384, SHA2_512) for "Encrypt-then-MAC" - NULL cipher with all MAC algorithms (CMAC, GMAC, XCBC_MAC, HMAC(SHA1, SHA2))
	HSE_SRV_ID_CRC32	Computes CRC32 checksum.
	HSE_SRV_ID_SIPHASH	SipHash is optimized for fast processing speeds when used to authenticate small messages. (MACs)
	HSE_SRV_ID_CMAC_WITH_COUNTER	Generates/verifies the CMAC of a given input message concatenated with a selected secure counter.
RNG	HSE_SRV_ID_GET_RANDOM_NUM	Get a random number. AIS31 and FIPS 140-2 compliant
Counters	HSE_SRV_ID_INCREMENT_COUNTER	Incrementing volatile counters. The Counter table can be published and load as an encrypted and authenticated blob using the HSE_SRV_ID_PUBLISH_LOAD_CNT_TBL service.
	HSE_SRV_ID_READ_COUNTER	Read volatile counter.
Advance Secure Booting (SMR/CR)	HSE_SRV_ID_SMR_ENTRY_INSTALL	Install a Secure Memory Region (SMR) table entry.
	HSE_SRV_ID_SMR_VERIFY	Verify (on demand) a Secure Memory Region (SMR) table entry.
	HSE_SRV_ID_CORE_RESET_ENTRY_INSTALL	Install a Core Reset (CR) table entry.
	HSE_SRV_ID_ON_DEMAND_CORE_RESET	On-demand release a core from a reset after loading and verification

Table 1. Available HSE Features/Services...continued

Service Class	HSE Service ID	Description/Notes
	HSE_SRV_ID_SMR_ENTRY_ERASE	Erase an SMR entry.
	HSE_SRV_ID_CORE_RESET_ENTRY_ERASE	Erase a CR entry.

7 Acronyms

Table 2. Acronyms

Acronym	Meaning / Description
ADKP	Application Debug Key / Password; one-time programmable parameter
AES	Advanced Encryption Standard; a cipher primitive
BCW	Boot Configuration Word
BAF	Boot Assist Flash
CUST_DEL	Device Life Cycle Customer Delivery
ECC	Error Correction Code
EL	Edge Lock
FA	Device Life Cycle Failure Analysis
FDK	Firmware Delivery Key
FW	Firmware
FXOSC	Fast External Crystal Oscillator
HBM	Host Boot Manager
HDR	Header
HMAC	Hash-based Message Authentication Code
HSE	Hardware Security Engine
ISO	International Organization for Standardization
IN_FIELD	Device Life Cycle is IN_FIELD
IVT	Image Vector Table
LC	Device Life Cycle; used to limit by design the configuration and debug / test possibilities of the device for in-field usage
MAC	Message Authentication Code
MCU_PROD	Device Life Cycle MCU Production
MDA	Master Domain Assignment
MRC	Memory Region Controller
MRGD	Memory Region Descriptor
MU	Messaging Unit
N/A	Not Applicable
NSK	NXP Signing Key
OEM_PROD	Device Life Cycle OEM Production
OTA	Over The Air
OTP	One Time Programmable
PDAC	Peripheral Domain Access Control
RNG	Random Number Generator
RSA	A public-key cryptosystem named after the inventors Mr. Rivest, Mr. Shamir and Mr. Adleman
SBAF/Secure BAF	Secure Boot Assist flash. Boot component in HSE.

Table 2. Acronyms...continued

Acronym	Meaning / Description
VT	Verification Tag

8 Revision History

Table 3. Revision history

Document ID	Release Date	Description
RN00275v.1	20 March 2025	Release notes for S32G3 HSE Firmware 0.2.64.0 SR Release

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